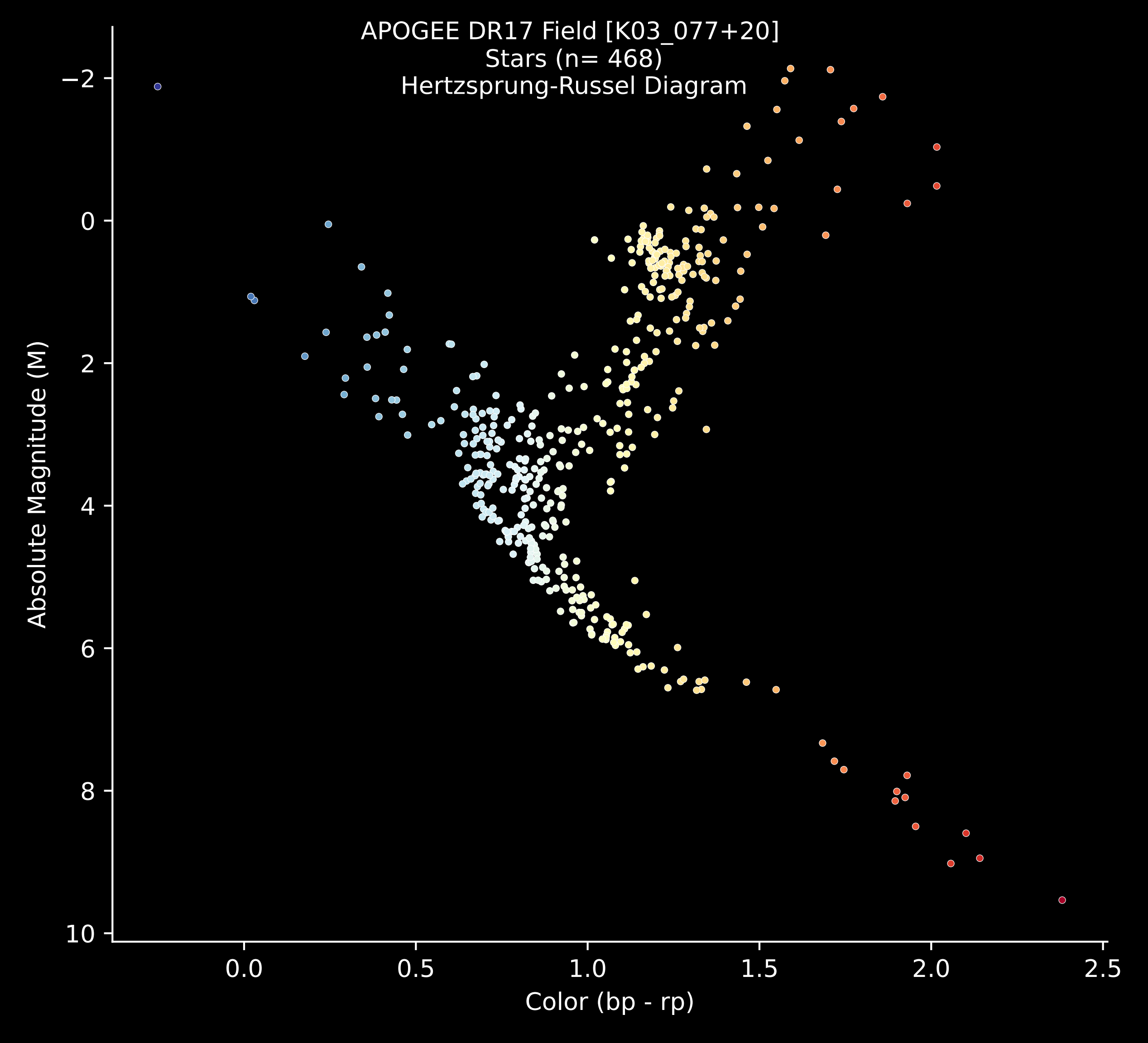
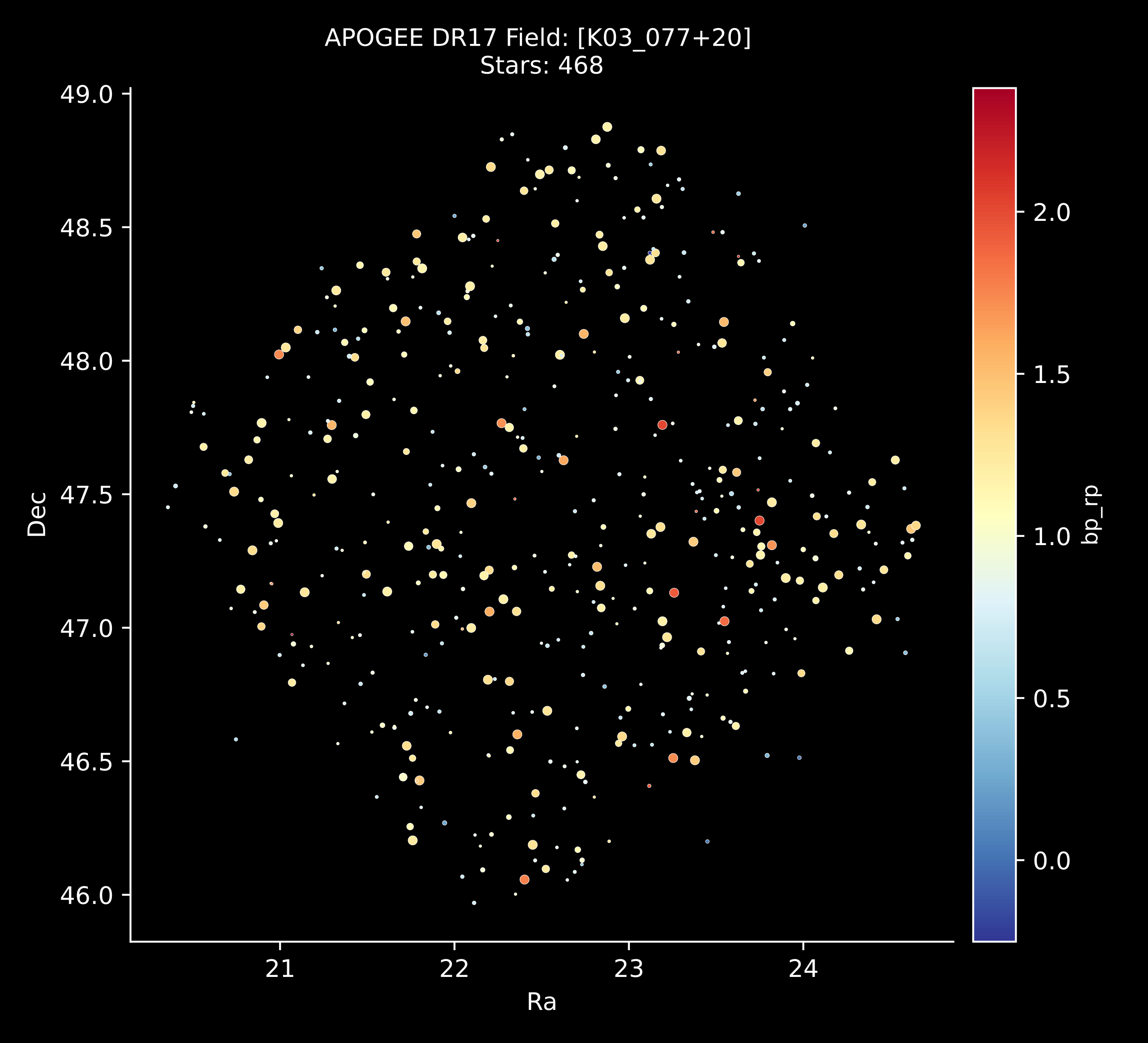
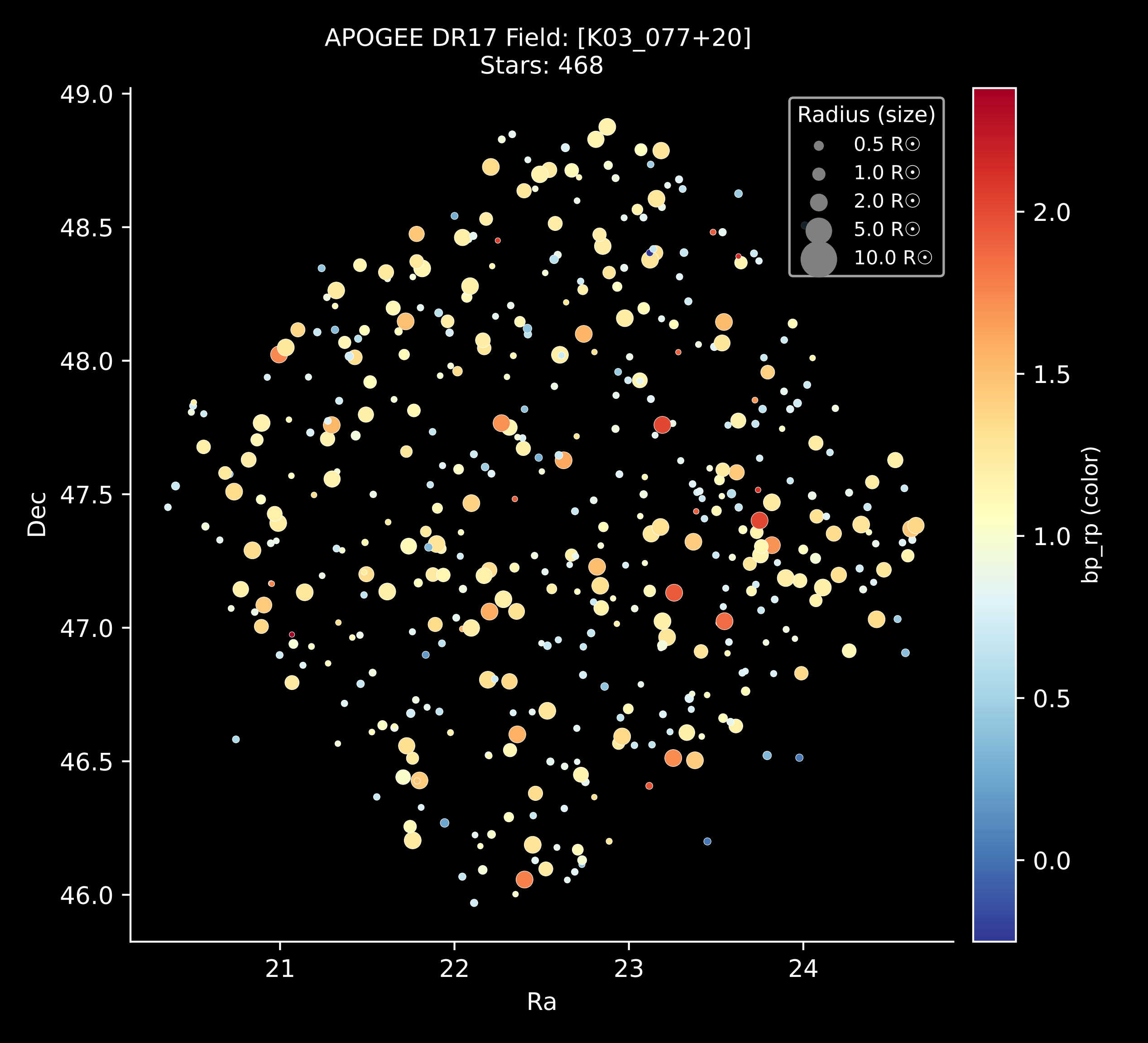
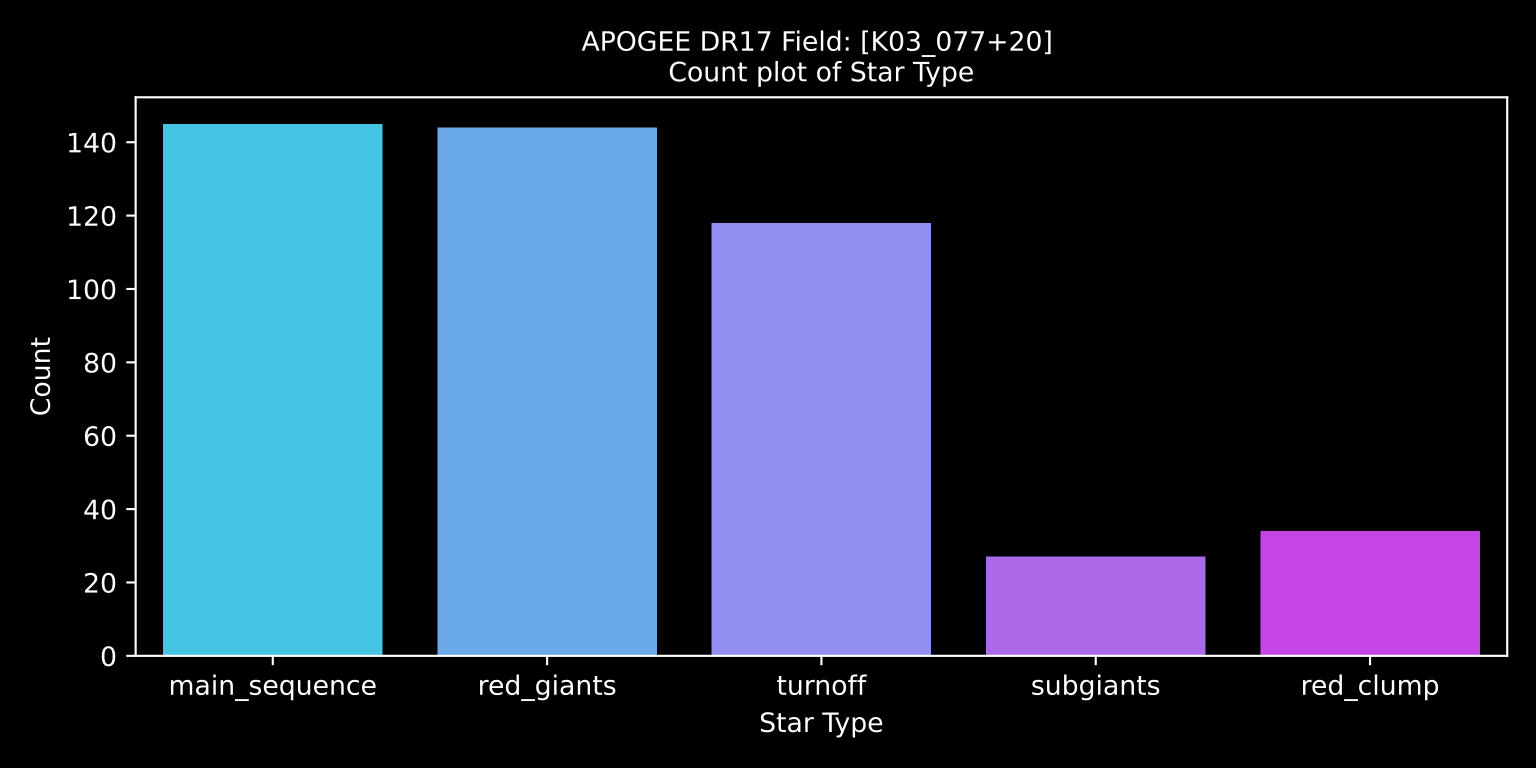
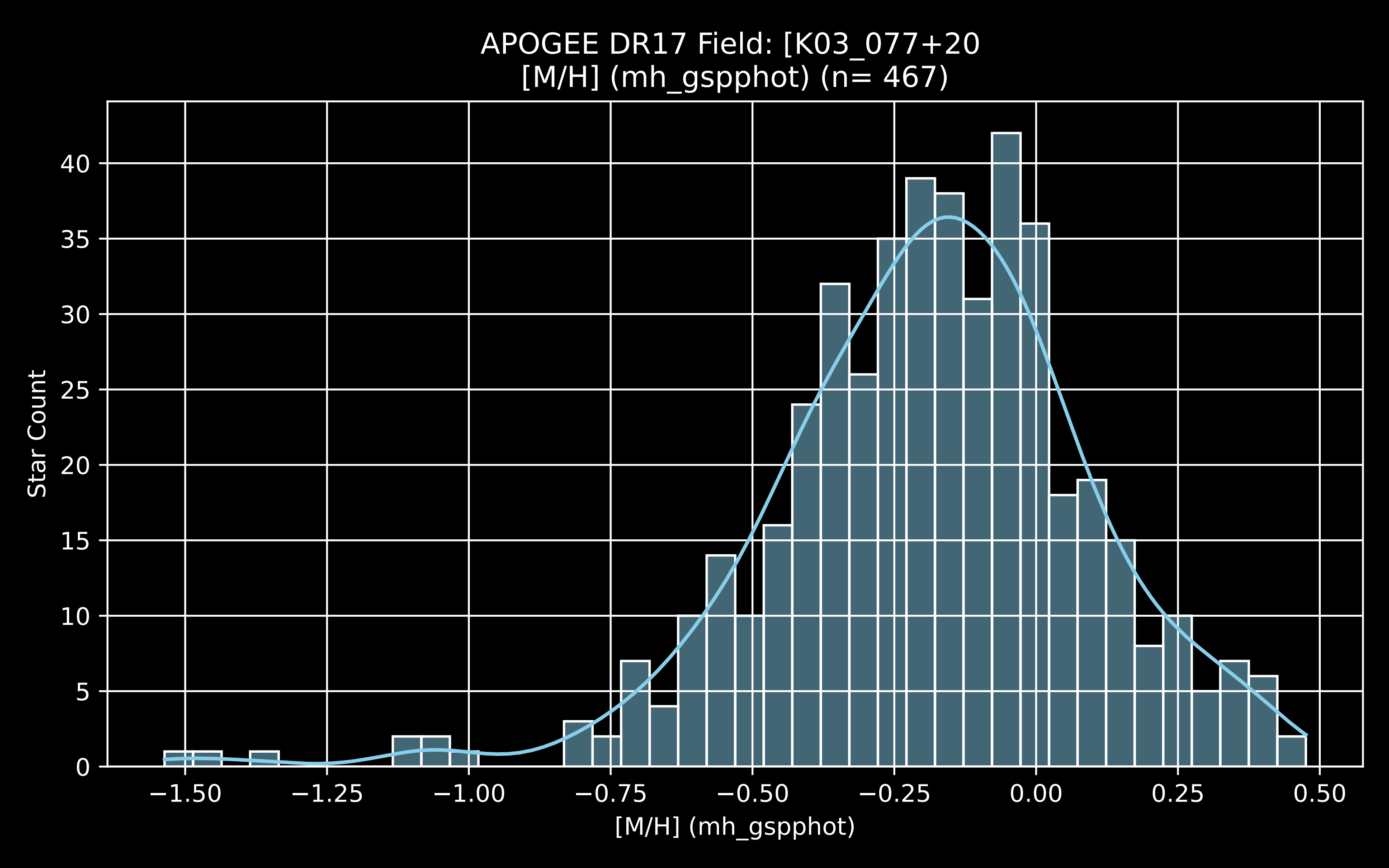


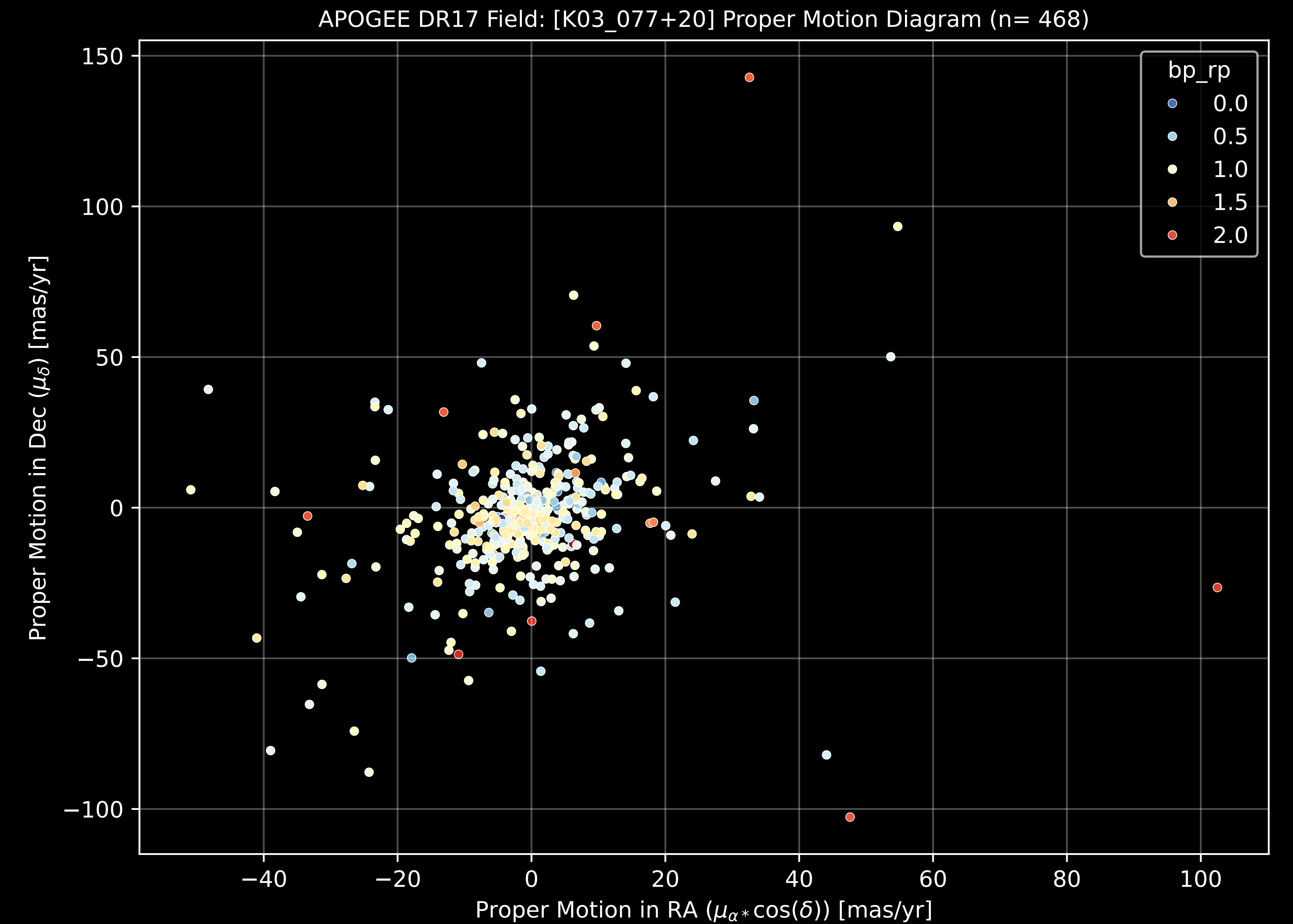
APOGEE DR17 Field [K03_077+20]
Stars (n= 468)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−2
0
2
4
6
8
10
0.0
0.5
1.0
1.5
2.0
2.5


APOGEE DR17 Field: [K03_077+20]
Stars: 468
Dec
Ra
bp_rp
49.0
48.5
48.0
47.5
47.0
46.5
46.0
21
22
23
24
2.0
1.5
1.0
0.5
0.0

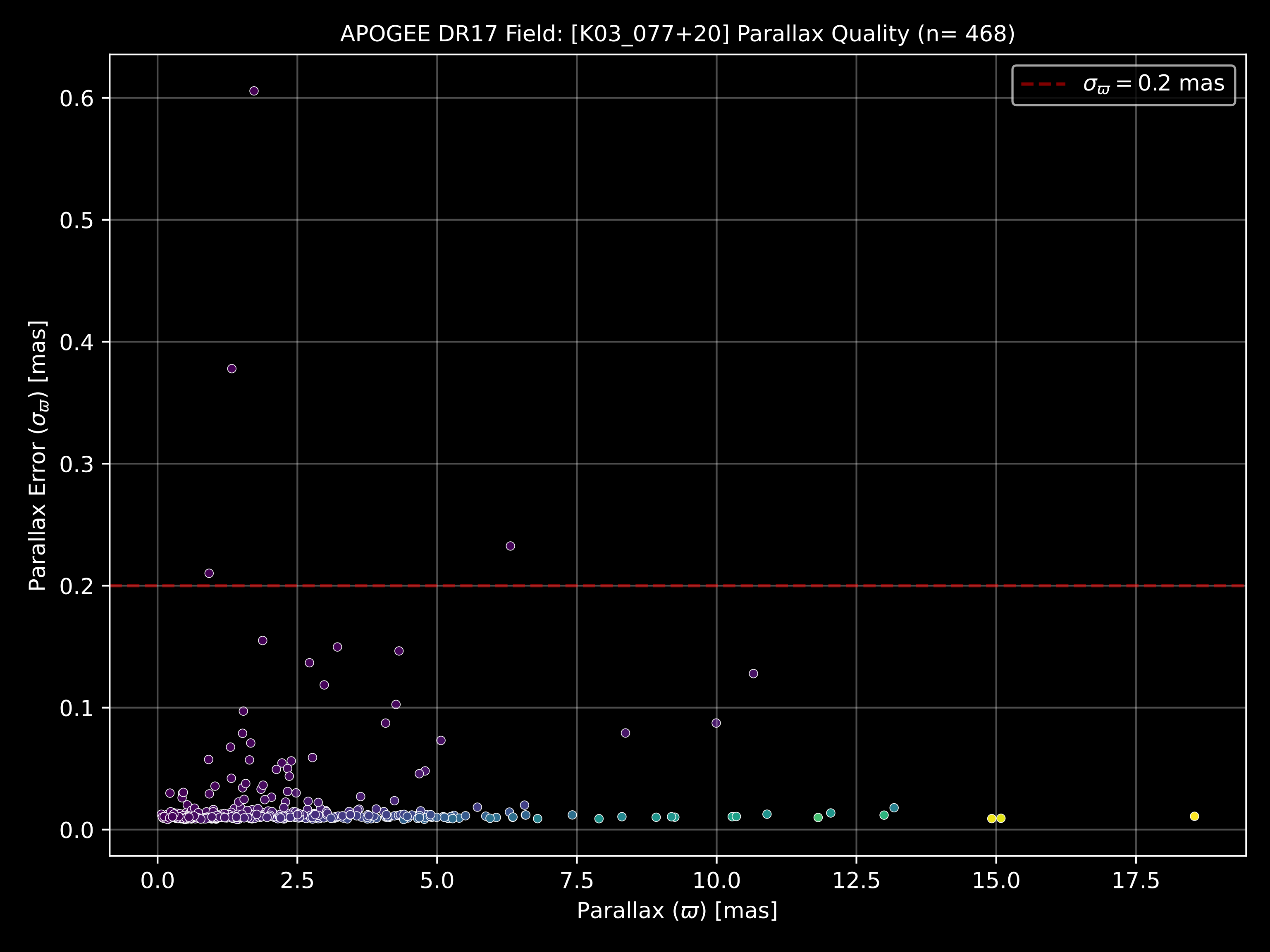

APOGEE DR17 Field: [K03_077+20]
Stars: 468
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
49.0
48.5
48.0
47.5
47.0
46.5
46.0
21
22
23
24
2.0
1.5
1.0
0.5
0.0


APOGEE DR17 Field: [K03_077+20]
Count plot of Star Type
Count
0
20
40
60
80
100
120
140
main_sequence
red_giants
turnoff
subgiants
red_clump
Star Type

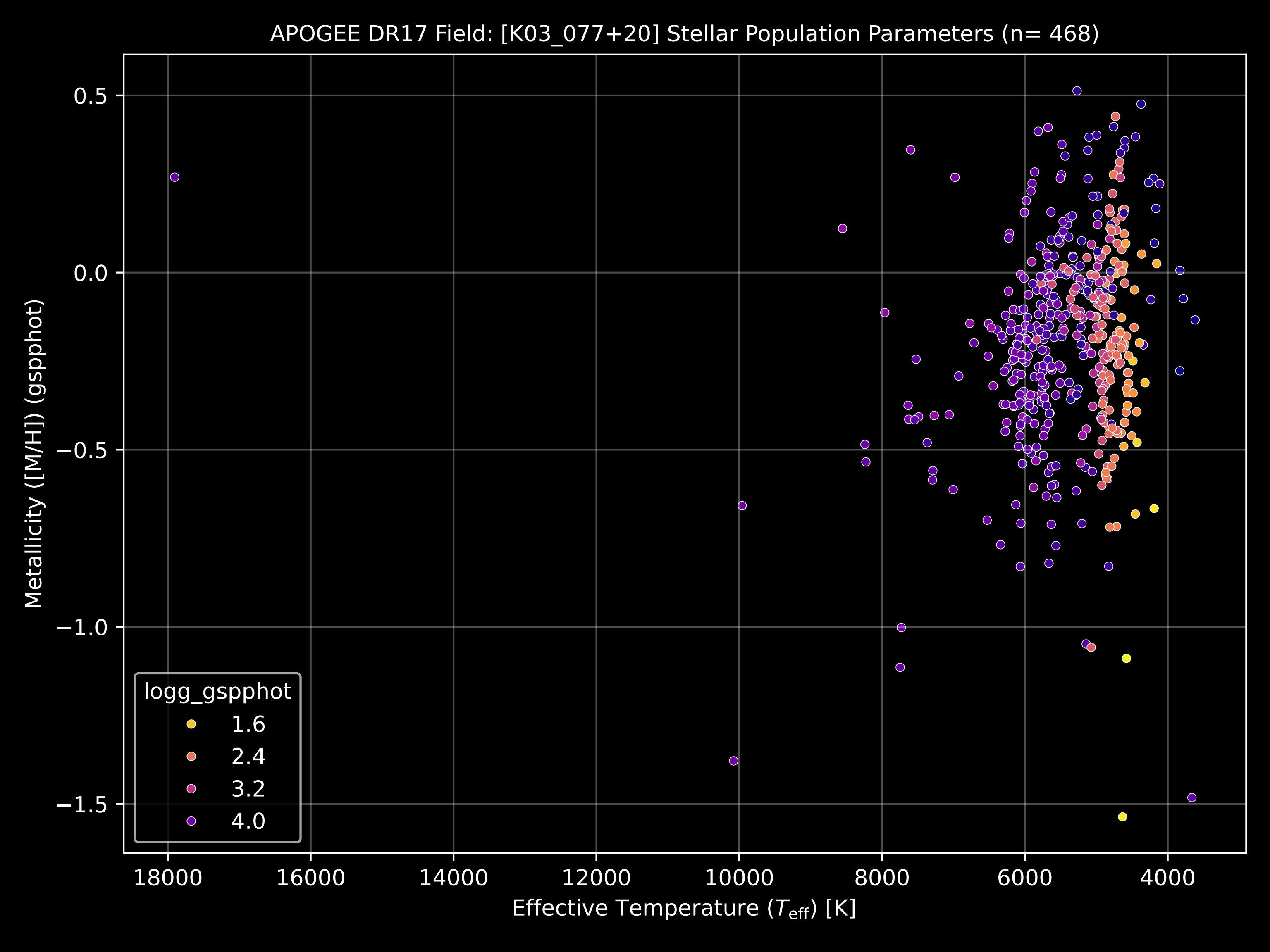

APOGEE DR17 Field: [K03_077+20
[M/H] (mh_gspphot) (n= 467)
Star Count
[M/H] (mh_gspphot)
0
5
10
15
20
25
30
35
40
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50

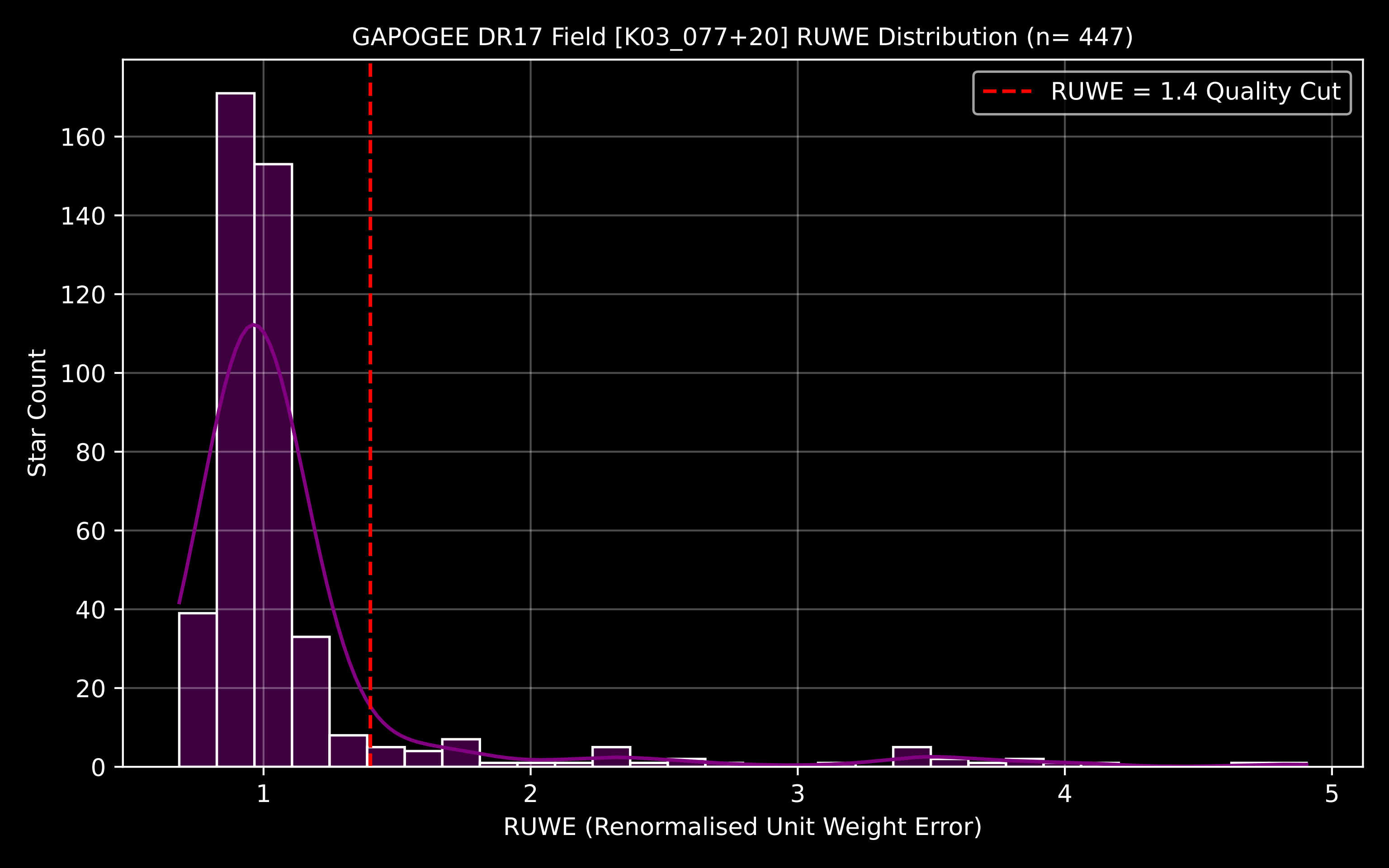


APOGEE DR17 Field: [K03_077+20] Proper Motion Diagram (n= 468)
bp_rp
0.0
0.5
1.0
1.5
2.0
150
100
50
0
−50
−100
−40
−20
0
20
40
60
80
100
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

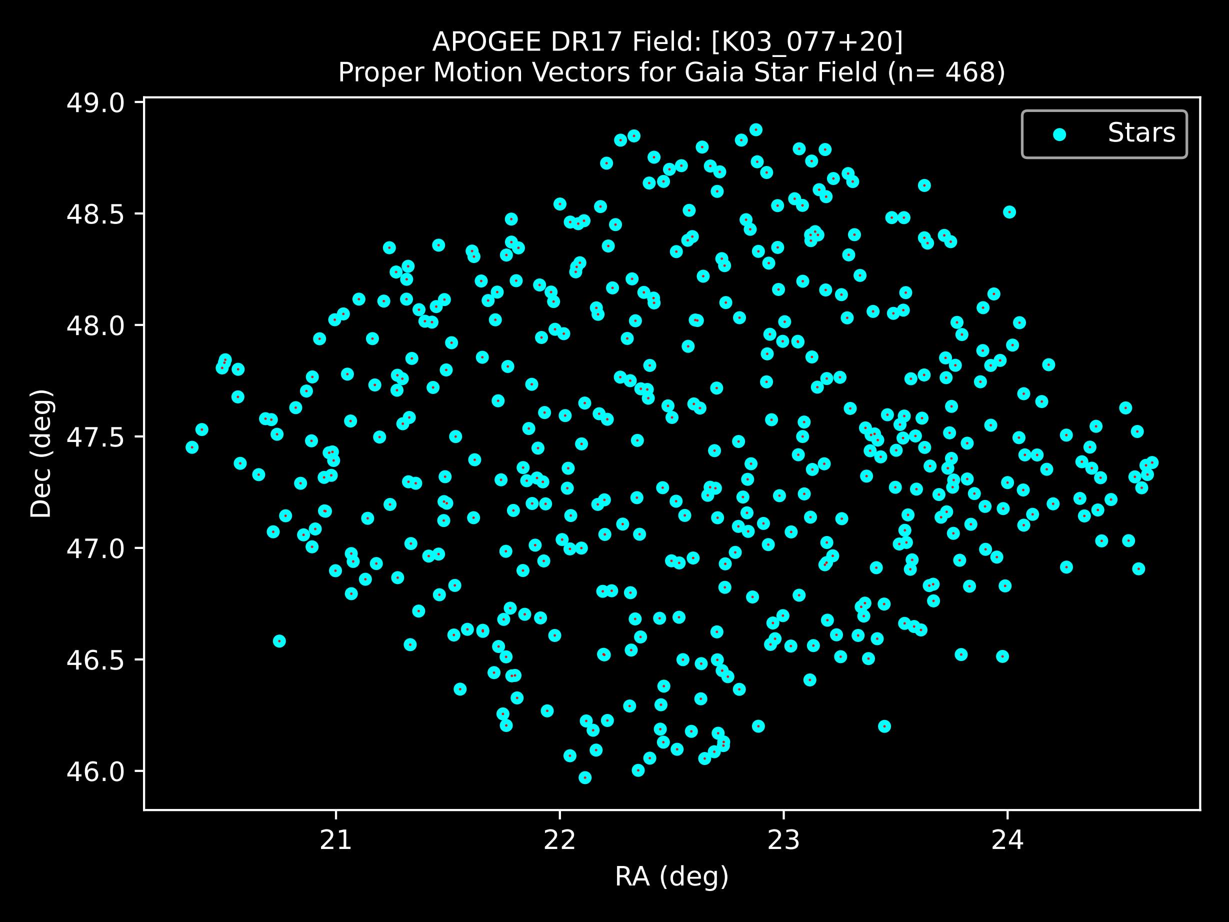


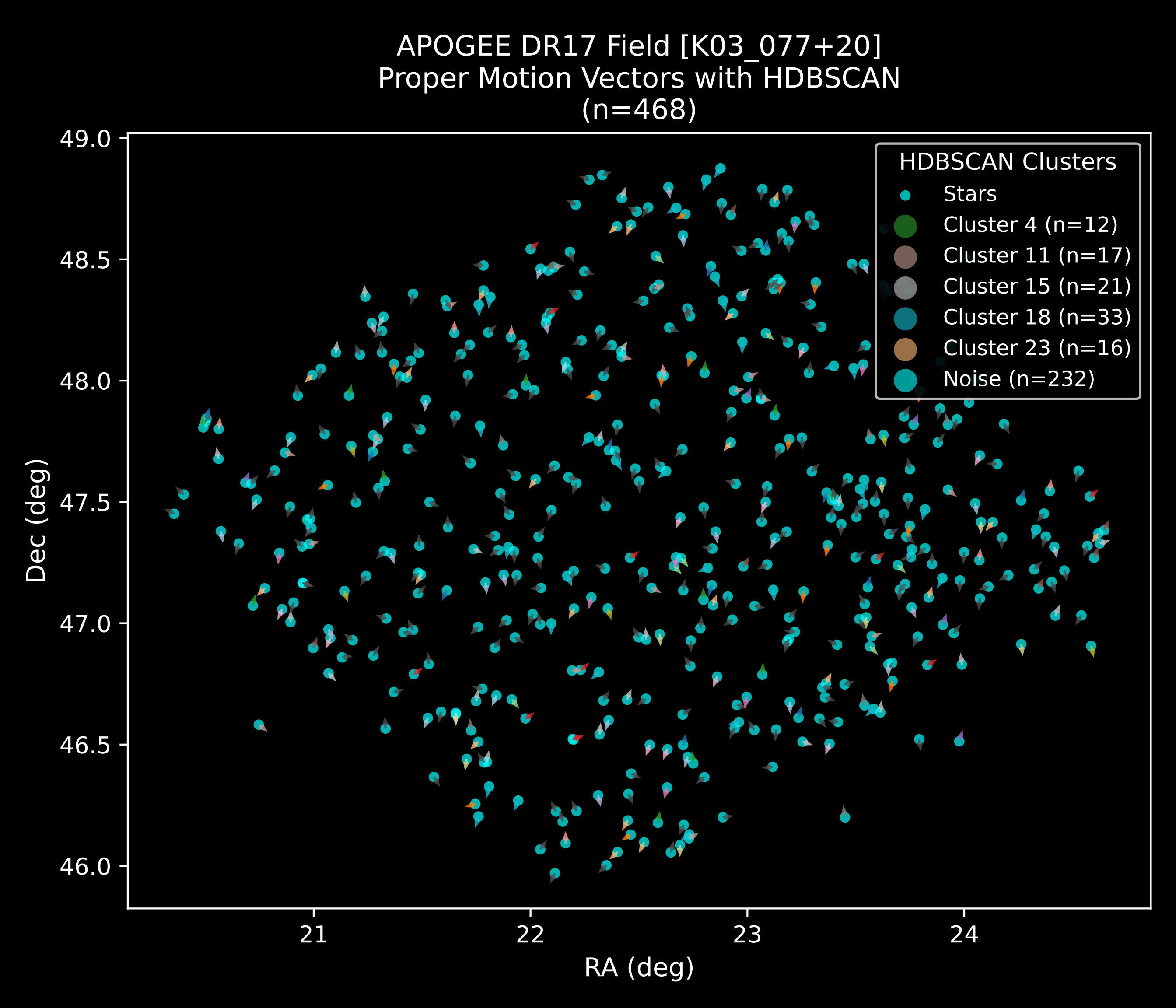
APOGEE DR17 Field: [K03_077+20] Parallax Quality (n= 468)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

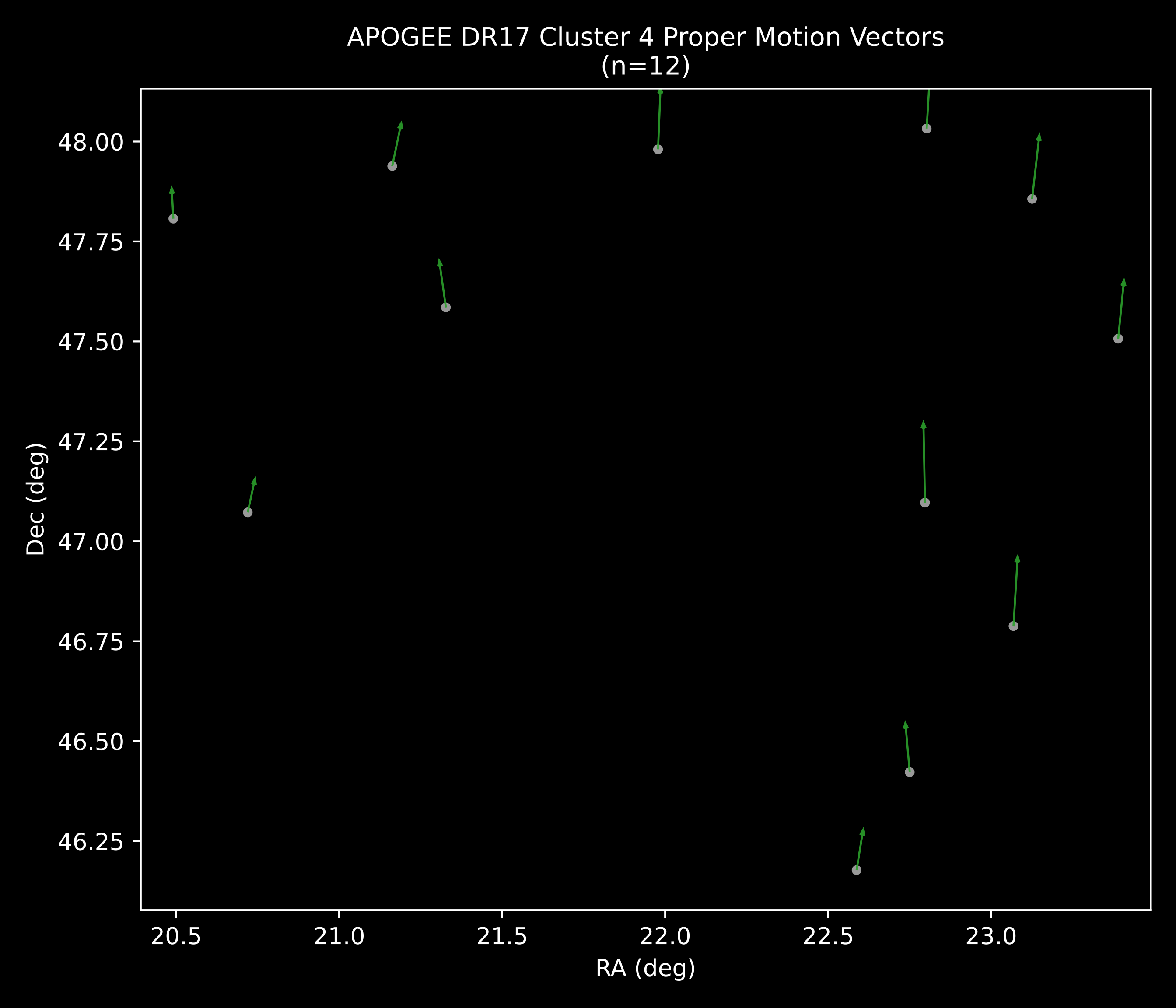

APOGEE DR17 Field: [K03_077+20] Stellar Population Parameters (n= 468)
Metallicity ([M/H]) (gspphot)
Effective Temperature (T_{eff}) [K]
logg_gspphot
1.6
2.4
3.2
4.0
0.5
0.0
−0.5
−1.0
−1.5
18000
16000
14000
12000
10000
8000
6000
4000

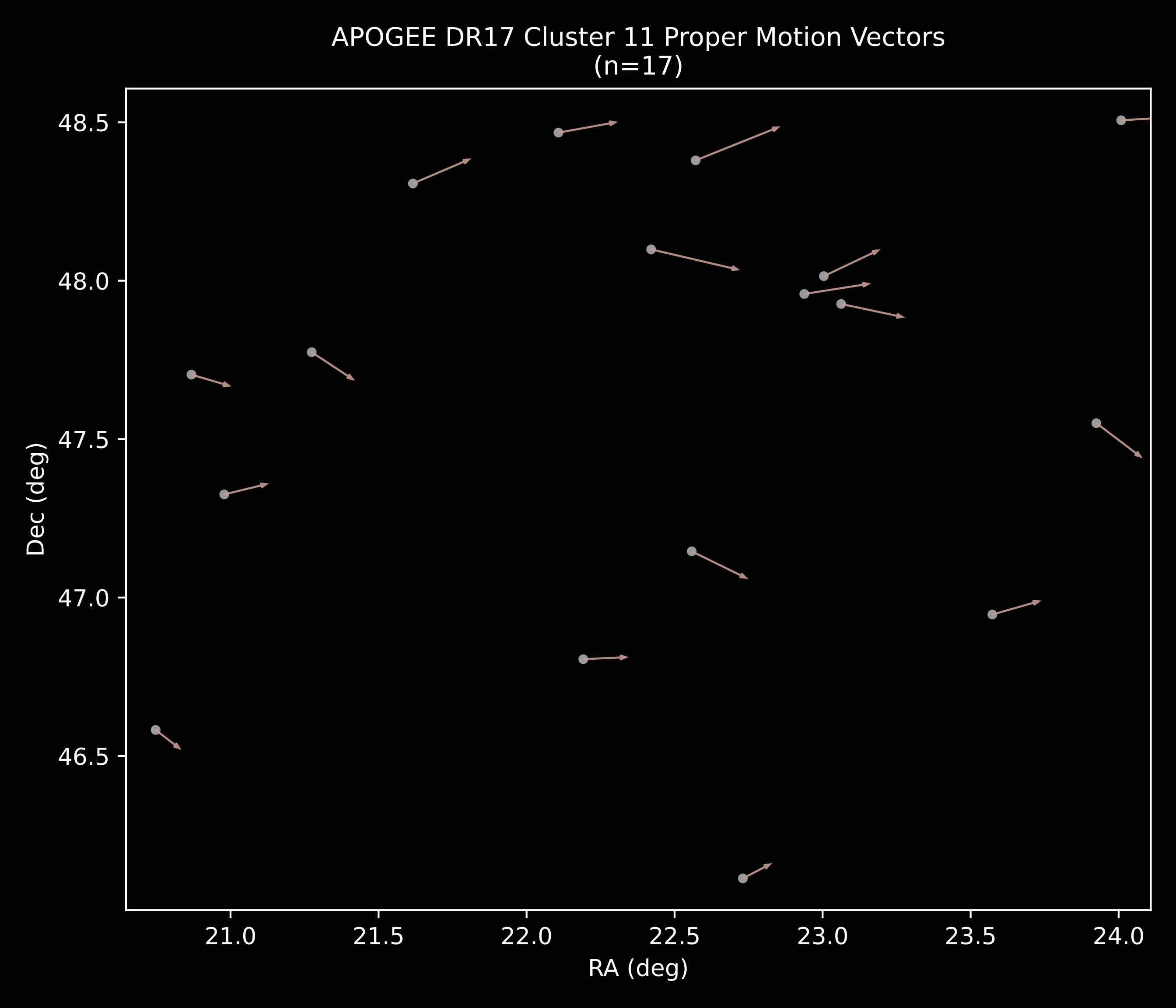


GAPOGEE DR17 Field [K03_077+20] RUWE Distribution (n= 447)
RUWE = 1.4 Quality Cut
Star Count
RUWE (Renormalised Unit Weight Error)
0
20
40
60
80
100
120
140
160
1
2
3
4
5

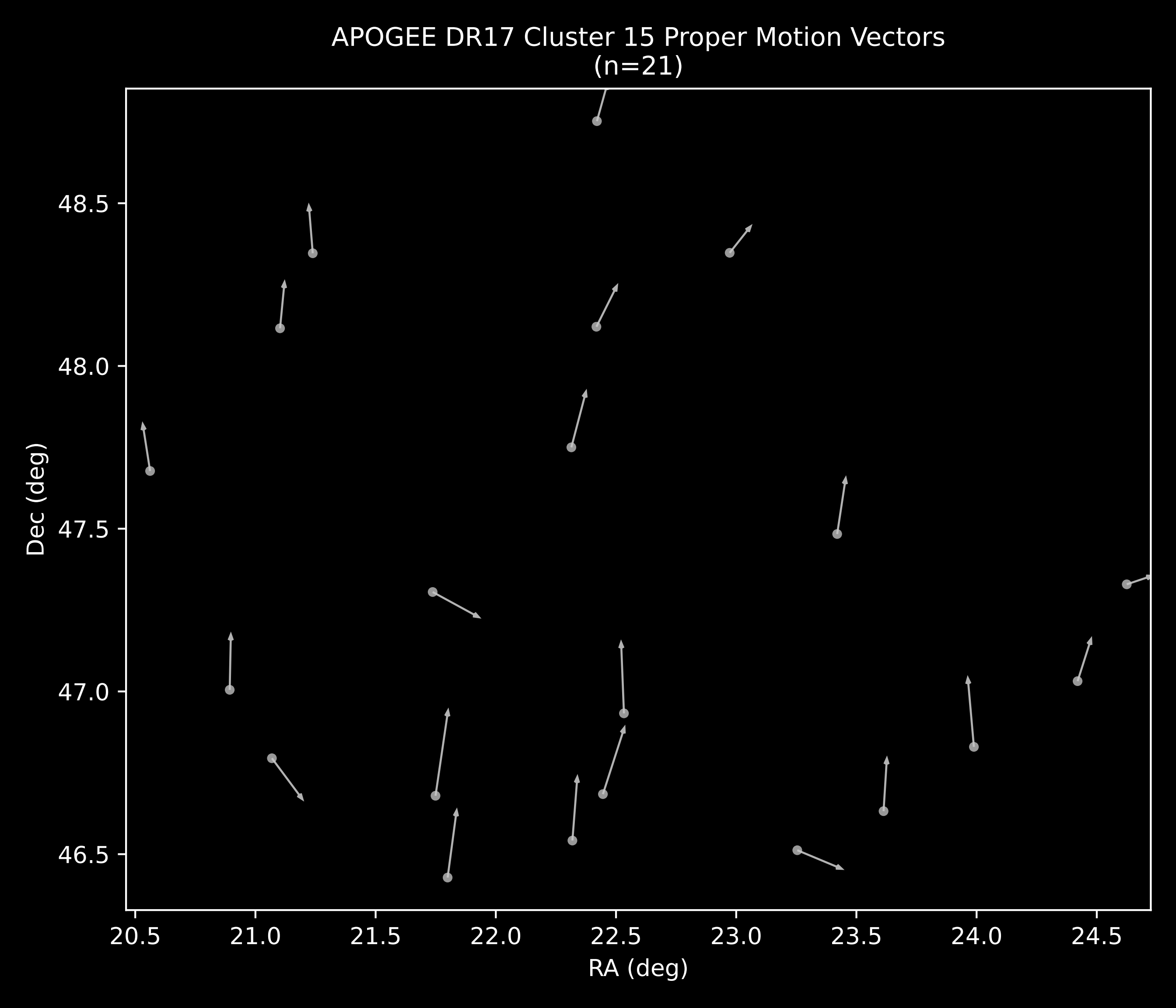

APOGEE DR17 Field: [K03_077+20]
Proper Motion Vectors for Gaia Star Field (n= 468)
Stars
Dec (deg)
RA (deg)
49.0
48.5
48.0
47.5
47.0
46.5
46.0
21
22
23
24

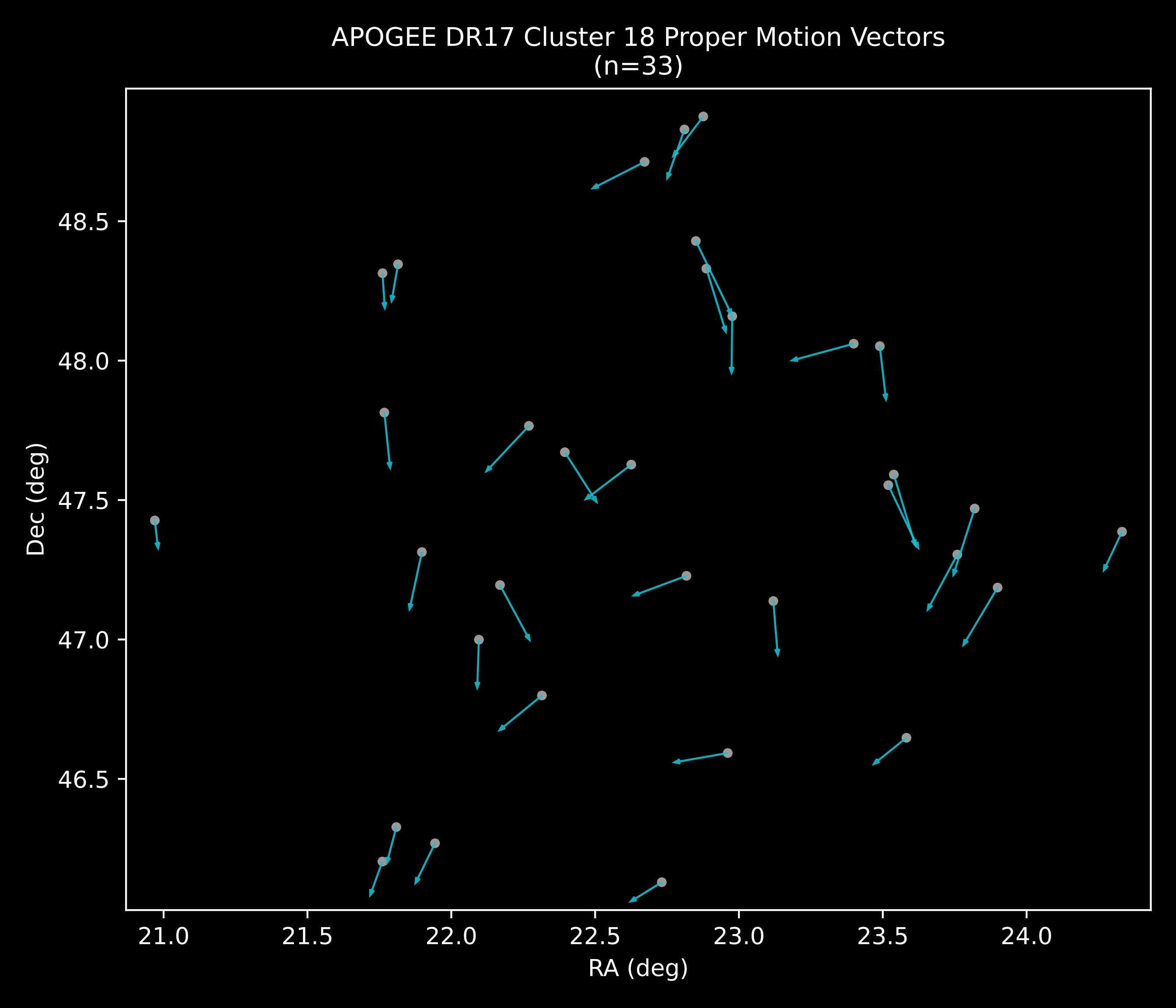

APOGEE DR17 Field [K03_077+20]
Proper Motion Vectors with HDBSCAN
(n=468)
HDBSCAN Clusters
Stars
Cluster 4 (n=12)
Cluster 11 (n=17)
Cluster 15 (n=21)
Cluster 18 (n=33)
Cluster 23 (n=16)
Noise (n=232)
Dec (deg)
RA (deg)
49.0
48.5
48.0
47.5
47.0
46.5
46.0
21
22
23
24

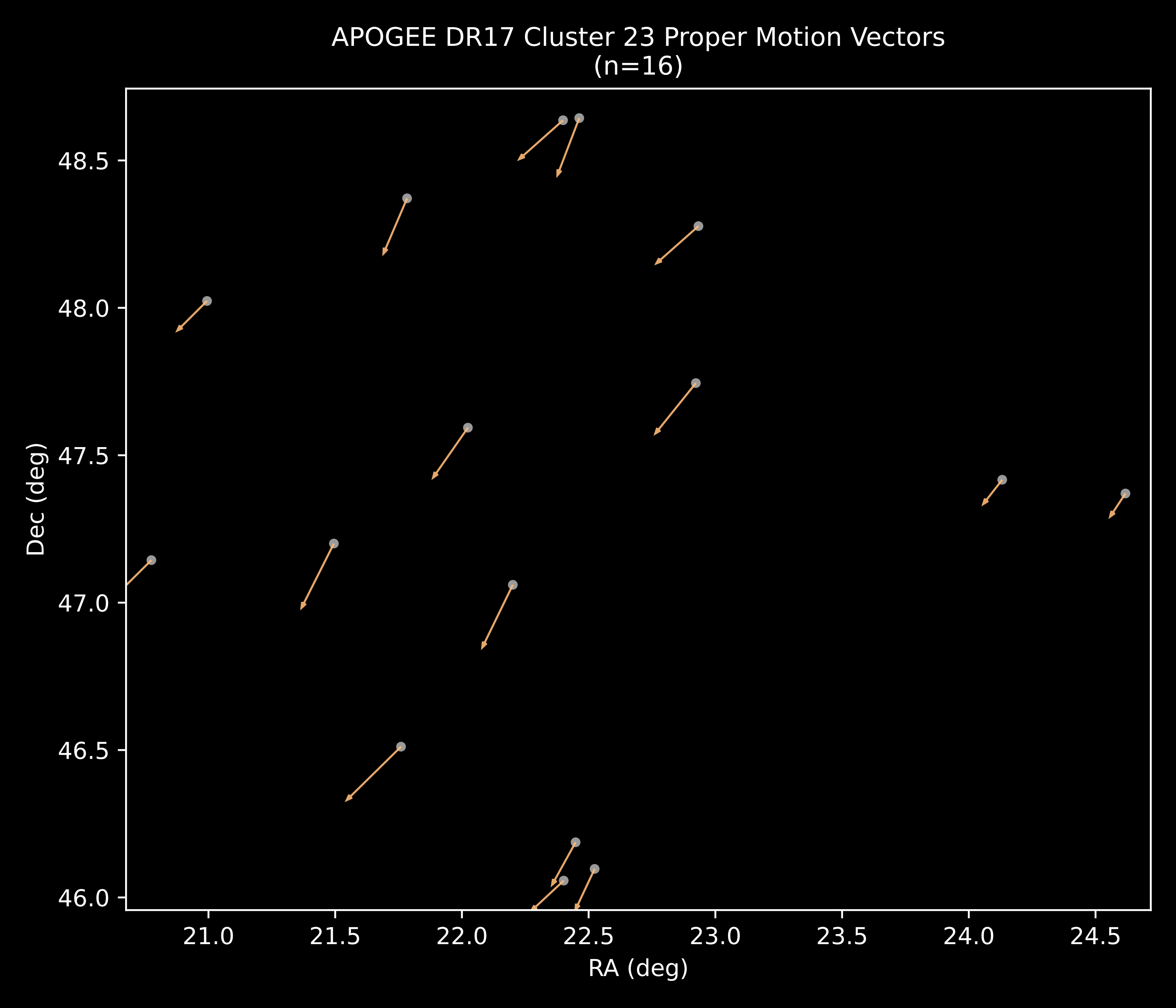

APOGEE DR17 Cluster 4 Proper Motion Vectors
(n=12)
48.00
47.75
47.50
47.25
47.00
46.75
46.50
46.25
Dec (deg)
20.5
21.0
21.5
22.0
22.5
23.0
RA (deg)

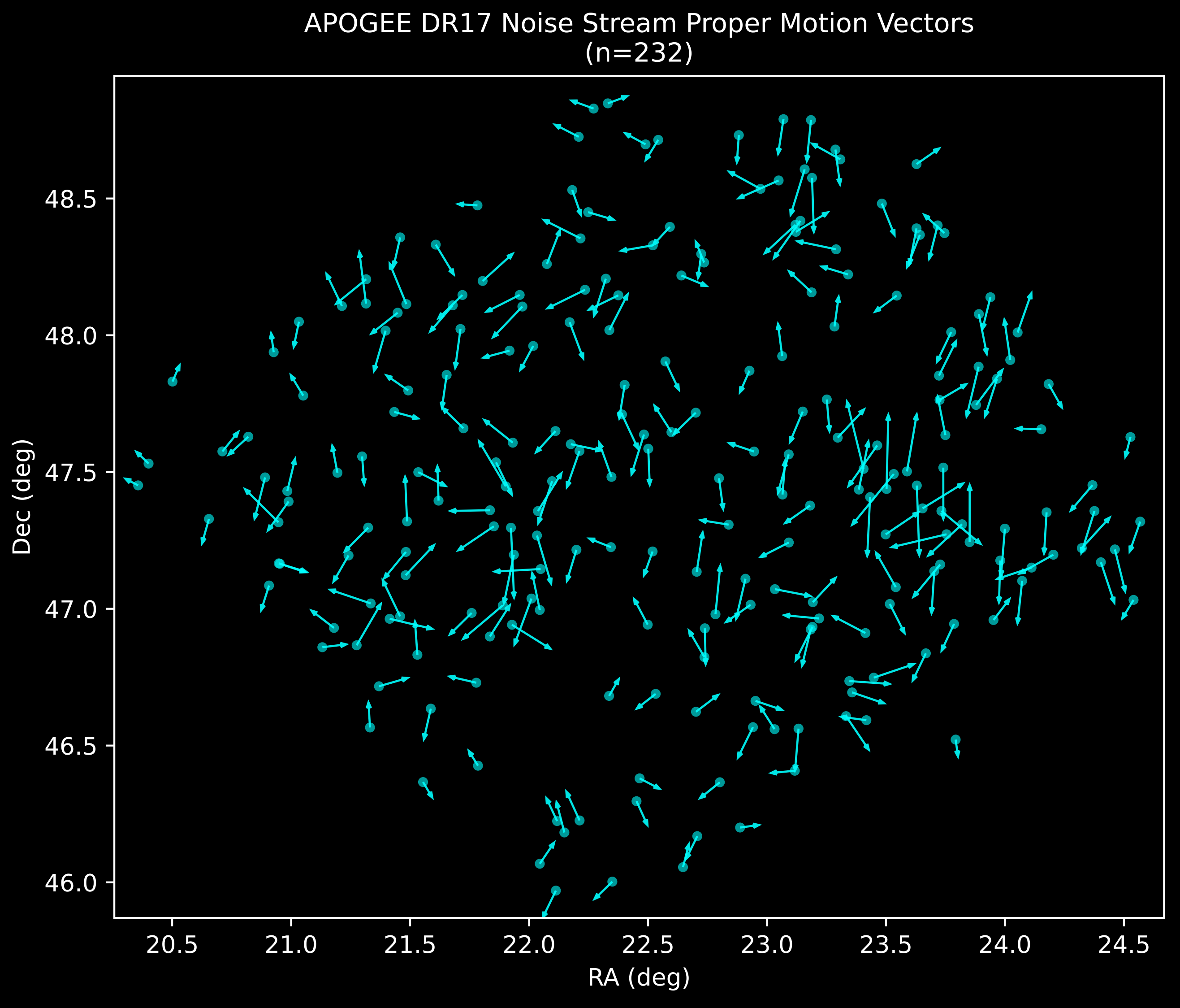


APOGEE DR17 Cluster 11 Proper Motion Vectors
(n=17)
48.5
48.0
47.5
47.0
46.5
Dec (deg)
21.0
21.5
22.0
22.5
23.0
23.5
24.0
RA (deg)


APOGEE DR17 Cluster 15 Proper Motion Vectors
(n=21)
48.5
48.0
47.5
47.0
46.5
Dec (deg)
20.5
21.0
21.5
22.0
22.5
23.0
23.5
24.0
24.5
RA (deg)


APOGEE DR17 Cluster 18 Proper Motion Vectors
(n=33)
Dec (deg)
RA (deg)
48.5
48.0
47.5
47.0
46.5
21.0
21.5
22.0
22.5
23.0
23.5
24.0


APOGEE DR17 Cluster 23 Proper Motion Vectors
(n=16)
Dec (deg)
RA (deg)
48.5
48.0
47.5
47.0
46.5
46.0
21.0
21.5
22.0
22.5
23.0
23.5
24.0
24.5


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=232)
Dec (deg)
RA (deg)
48.5
48.0
47.5
47.0
46.5
46.0
20.5
21.0
21.5
22.0
22.5
23.0
23.5
24.0
24.5